

| Question |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks available |     |     |       |       |      |
|----------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 2        | (a) |  | At least 4 field lines are radial and symmetrical (1)<br>At least 2 equipotentials are circles / spheres (1)<br>Out of positron and into anti-neutron (1)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3               |     |     | 3     |       |      |
|          | (b) |  | <b>Indicative content:</b><br><br><b>Differences</b><br>1. electric fields - charges & grav fields – masses (or force per unit mass, force per unit charge or forces between charges & masses)<br>2. e-fields – into & out, g-fields only into (direction point – also accept grav forces only attractive, e-forces attractive & repulsive)<br>3. Gravity is far weaker (on small scale) OR electrical negligible on large scale (i.e. scale point)<br>4. E potential work done per unit charge & G work done per unit mass<br>5. E-fields depend on permittivity OR g-fields depend on dark matter<br><br><b>Similarities</b><br>1. Same shape field (for point/spherical masses/charges). Note: this point is <b>different</b> from the <b>same shape graph</b> which is the next point.<br>2. inverse square law (applies to both)<br>3. Both fields lead to (concept of) PE<br>4. Potential or PE is inversely proportional to distance<br>5. Both fundamental force laws<br>6. Both have scalar fields OR both have vector fields (or both!)<br>7. Anything to do with exchange particles (photon, gluon, bosons)<br><br>NOTE: some differences can be viewed as similarities and vice versa e.g. energy per unit charge and energy per unit mass could be considered a similarity. Also, the different exchange particles could be considered a difference. | 6               |     |     | 6     |       |      |

| Question |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks available |     |     |       |       |      |
|----------|--|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  |  | <p><b>5-6 marks</b><br/>Similarities and differences between electric and gravitational fields covered comprehensively.<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i></p> <p><b>3-4 marks</b><br/>Similarities or differences between electric and gravitational fields covered comprehensively or limited description of both.<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i></p> <p><b>1-2 marks</b><br/>Limited description of either the similarities or the differences.<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i></p> <p><b>0 marks</b><br/>No attempt made or no response worthy of credit.</p> |                 |     |     |       |       |      |
|          |  |  | Question 2 total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 9               | 0   | 0   | 9     | 0     | 0    |